6 A nichrome wire of cross-sectional area $5.7 \times 10^{-8}\,m^2$ is connected to a power supply. The potential difference (p.d.) across the wire is 230 V. The power dissipated in the wire is 1.0 kW.

(a) Show that the resistance of the wire is $53\,\Omega$.

[2]

(b) The resistivity of nichrome is $1.3 \times 10^{-6}\,\Omega\,m$.

Calculate the length of the wire.

length = ... m [2]

(c) The average drift speed of the charge carriers (free electrons) in the nichrome wire is $5.4 \times 10^{-3}\,m\,s^{-1}$.

Calculate the number density of charge carriers of nichrome.

number density = ... m^{-3} [3]

[Total: 7]